

# Mohannad Darandari

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## Summary

Artificial Intelligence Engineer specializing in building production-ready AI systems, LLM-powered applications, and intelligent automation solutions. Experienced in developing AI agents, integrating APIs, designing scalable backend systems, and transforming complex data into actionable intelligence. Focused on delivering high-impact AI solutions that improve efficiency, automate workflows, and solve real-world business challenges.

## Skills

- **Skills:** Python, Java, C/C++, Oracle SQL (PL/SQL), Machine Learning, Deep Learning, Large Language Models (LLMs), AI Agents Development, Retrieval-Augmented Generation (RAG), Prompt Engineering, LangChain, Vector Databases (FAISS, Pinecone, Chroma), Robotic Process Automation (RPA), Workflow Automation (n8n, Power Automate, UiPath, Automation Anywhere), API Design & Integration, System Integration, Backend Development, Data Visualization (Power BI, Tableau), Database Automation & Optimization, Docker, Multi-agent Systems, Git & GitHub, Linux

## Education

### University of Jeddah | College of Computer Science & Engineering

**Dec 2020 - Dec 2024**

*Bachelor's, Artificial Intelligence Engineering*

*Jeddah, Saudi Arabia*

- **Coursework:** Machine Learning, Deep Learning, Neural Networks, Computer Vision, Natural Language Processing (NLP), Reinforcement Learning, Artificial Intelligence, Advanced Algorithms, Data Structures, Probabilistic Models, Pattern Recognition, Optimization Techniques, Robotics & Autonomous Systems, Data Mining, Statistical Learning, Distributed Systems for AI, Python for AI, AI Ethics.

### Southern Illinois University

**Oct 2019 - Nov 2020**

*ESL, English as a Second Language*

*Carbondale, United States of America*

### DePaul University

**Sep 2019 - Oct 2020**

*ESL, English as a Second Language*

*Chicago, United States of America*

## Work Experience

### BTC Networks | AI & Automation Engineer

**Aug 2025**

- Designed and deployed a full internal MCP-based AI system and cross-platform application (desktop & mobile) for BTC employees, featuring an on-prem local LLM trained on Oracle enterprise data, RAG pipelines, and advanced NL2SQL with dynamic multi-table SQL query generation, automatically generating interactive sales dashboards and charts from natural-language queries to deliver secure, real-time, role-based business insights with zero direct database exposure.
- Led the full design and development of BTC's corporate website from scratch using React, and built an AI-powered chatbot inside the website integrated with n8n, Power Automate, Outlook, and Google Sheets to automate HR and Sales workflows, including real-time client interaction, CV collection and routing, intelligent lead handling, client appointment reservations, and seamless escalation to support teams.
- Built the BTC Contracts Health with AI dashboard, applying predictive analytics and machine learning in Power BI to monitor budgets, forecast deviations, and assign risk scores, helping management make proactive financial decisions.
- Developed a Cybersecurity machine learning model analyzing 40,000+ security logs across 25 features, detecting anomalies and patterns to strengthen BTC's threat monitoring and proactive defense capabilities.

### MASHARIQ - Contract | Business Intelligence Consultant

**May 2025 - Jul 2025**

- Built and automated real-time Power BI dashboards to monitor the performance of Hajj and Umrah service centers, tracking metrics such as participant attendance, evaluation submissions, and company-specific KPIs (Rakeen & Sanaa).
- Integrated multiple data sources (Google Sheets, Excel, and cloud databases) to streamline daily reporting for field teams during Hajj season, reducing manual data entry and errors by over 99.9%.
- Automated alerts and performance summaries in Power BI to flag underperforming centers and trigger follow-up actions, enhancing service quality and responsiveness.

### AlFares Trading Co. | IT Specialist

**Apr 2023 - Apr 2025**

- Led a team in developing automation and machine learning projects at AlFares Trading Co.
- Implemented innovative solutions to streamline processes and improve efficiency.
- Successfully integrated machine learning algorithms to optimize company operations.
- Collaborated with cross-functional teams to drive project success.

## Internships & Bootcamps

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### Nahdi Medical Co.

Jul 2024 - Dec 2024

#### *Supply Chain Excellence*

- Designed and developed interactive Microsoft Power BI dashboard for real-time insights, enhancing decision-making.
- Automated strategic plan execution involving 260+ tasks annually, achieving significant time savings.
- Collaborated with cross-functional teams to integrate AI-driven automation into daily operations, supporting digital transformation initiatives.

### Smart Methods

Jun 2024 - Jul 2024

#### *Artificial Intelligence and Robotics Operating Systems (ROS) Full Stack Engineering*

- Designed and programmed robotic legs, robotic arm control, HuskyLens integration with Arduino, ROS simulation, and building a bridge between ROS and ROS2 for advanced robotics compatibility

### KAUST (King Abdullah University of Science and Technology)

Jan 2024 - Feb 2024

#### *Introduction & Advanced AI Program*

- Applied advanced data preprocessing, model training, and evaluation techniques on large datasets, with focus on performance tuning and model interpretability.
- Collaborated with a team of AI engineers and data scientists on innovation challenges aligned with Saudi Vision 2030, gaining exposure to AI applications in healthcare, energy, and sustainability.
- Completed a competitive, hands-on training program focused on cutting-edge AI techniques including machine learning, deep learning, computer vision, and NLP, delivered by experts from KAUST and international partners

### SDAIA (Saudi Data & AI Authority)

Dec 2023 - Jan 2024

#### *Machine Learning Bootcamp*

- Trained and deployed machine learning models using Python, focusing on predictive analytics and decision-making.
- Built data pipelines to clean, preprocess, and prepare data for model training, achieving high accuracy rates in classification and regression tasks. Used data visualization tools to generate insights from complex datasets
- Conducted a group project that implemented supervised learning algorithms to solve a real-world problem, presenting the findings to stakeholders. Gained hands-on experience in model evaluation techniques, including model precision-recall analysis and hyperparameter tuning.

### MCIT (Ministry of Communications and Information Technology of Saudi Arabia)

Dec 2022 - Mar 2023

#### *Robotic Process Automation (RPA) Bootcamp*

- Completed a 400-hour government-sponsored RPA bootcamp focused on enterprise automation and process optimization.
- Designed and implemented automated workflows using UiPath and Automation Anywhere, reducing processing time.
- Developed custom RPA bots to handle complex, multi-step processes such as invoice validation, task scheduling, and data entry across systems. Enhanced operational efficiency by integrating automation tools with enterprise applications, ensuring seamless task execution.

### SDAIA (Saudi Data & AI Authority)

Sep 2022 - Oct 2022

#### *Artificial Intelligence Foundation Bootcamp*

- Implemented Deep Learning models using TensorFlow and Keras to optimize operational workflows and predictive analytics
- Utilized Machine Learning techniques to analyze complex datasets, providing actionable insights to support data-driven decision-making
- Applied Artificial Neural Networks (ANNs) and Statistical methods to create scalable algorithms for real-time data processing and trend forecasting.

## Projects

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### Enterprise MCP AI Assistant with RAG, Query Generation, and Role-Based Access Control | *BTC Networks*

- Designed and deployed a full MCP AI platform powered by a local LLM integrated with Oracle databases to enable secure, on-premise intelligent operations.
- Enabled employees to query SOI and ISA Oracle tables, retrieve employee data, and automatically generate SQL queries using natural language.
- Built RAG pipelines to search internal Oracle-connected company documents and knowledge bases with high accuracy.
- Developed MCP tools for automated chart generation, executive dashboards, and real-time Oracle analytics, empowering employees and C-level executives with instant, actionable insights.
- Implemented secure SSO authentication and role-based access control (RBAC) to ensure enterprise-level security and controlled data access.

### Automation Project the Supply Chain Strategy | *Al Nahdi Medical Co.*

- Developed a comprehensive automation cycle for managing a 3-year strategic plan comprising 260+ tasks and projects annually. Built and implemented 9 custom bots using Power Automate, ensuring real-time synchronization between Excel, Planner, Outlook, Power automate and Power BI. Delivered daily task updates and automated email reminders for the assignees in the strategy, accurate follow-up without errors and significantly improving efficiency and collaboration.

## **AI-Powered Glioblastoma Detection and Prediction System | *University of Jeddah***

- Designed an advanced medical AI clinical decision support system for detecting and predicting brain tumors using multimodal radiology data (MRI, CT, and genomic images), achieving 97% accuracy and assisting physicians in early diagnosis and treatment planning.
- Built an automated deep learning model for precise glioblastoma detection and early diagnosis.
- Utilized innovative image analysis techniques to improve diagnostic accuracy and enhance treatment planning.
- Developed a user-friendly interface for clinicians to upload medical images and receive diagnostic results.
- Integrated the solution into clinical workflows, streamlining processes and improving patient outcomes.
- Presented the project to healthcare stakeholders, emphasizing transformative impact on diagnostic practices.

## **Refund Process Automation at Gyms | *MCIT***

- Created a workflow using Automation Anywhere to automate the refund process for gym memberships. The system efficiently handled email communication, customer data storage, and refund approvals, reducing manual errors by 99.9% and processing time by 80%. Scaled the solution for multi-location gyms.

## **Certifications**

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- Generative AI | AWS | 2025
- IEEE Member (Computational Intelligence Society, Biometrics Council) | IEEE | 2025
- Certified Supply Chain Analyst (CSCA) | ISCEA | 2025
- Certified Full Stack Robotic Engineer | Smart Methods | 2024
- Certified Machine Learning Practitioner | SDAIA | 2024
- Advanced Artificial Intelligence Certificate | KAUST | 2024
- Certified Advanced Robotics Process Automation Professional (RPA) | Automation Anywhere | 2023
- Red Belt in UiPath | Automation Anywhere | MCIT | 2023
- Career Ready Certification | Essentials Program | MBS Foundation | 2023
- Google Data Analytics Professional Certificate | Google | 2022
- Artificial Intelligence Specialization | IBM | 2022
- Chatbot Building | IBM | 2022
- Cybersecurity Essentials | Cisco Academy | 2022

## **Languages**

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- Arabic (Native)
- English (Professional)